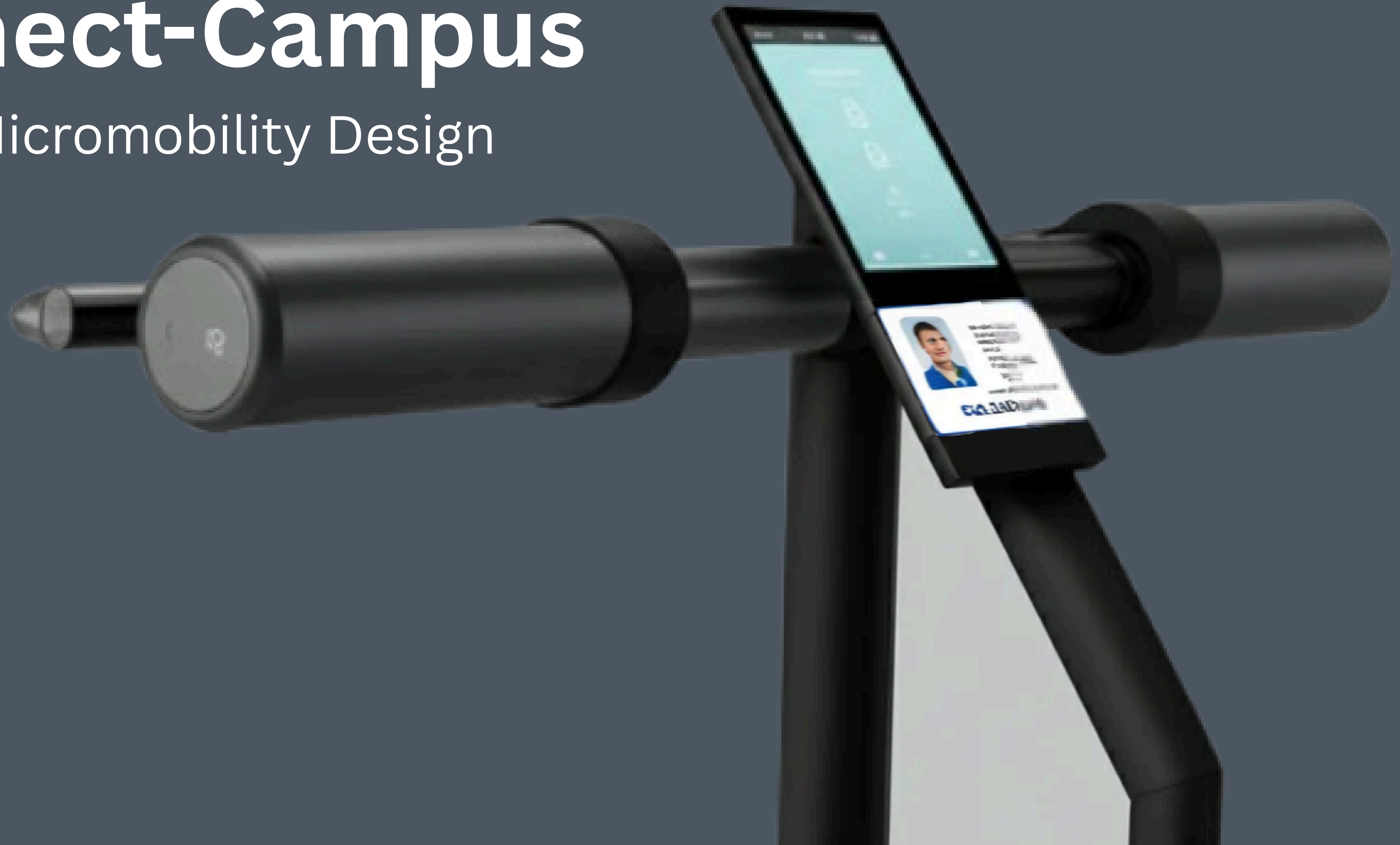


Project: Connect-Campus

A Systems-Based in Micromobility Design



Sri Charan Reddy

The On-Site "Last Mile"

Large campuses are small cities, but they lack smart, on-demand infrastructure. This project proposes an integrated ecosystem to solve the on-site mobility gap.

My Role & Process



Role: Lead Product Designer (Solo Project)



Responsibilities: User Research (Qual/Quant), Market Analysis, Industrial Design (Vehicle/Hub), CMF Strategy, Manufacturing & Cost Analysis.



Process: This case study follows a 4D process: Discover, Define, Develop, and Deliver.

1. Discover & Define

Validating the problem with data and empathy.

Research Methodology

-  **Secondary Research:** Market scan of existing public (Bird, Lime) and private (shuttle) "solutions" to identify feature gaps and business failures.
-  **Primary (Qualitative):** 1-on-1 interviews (n=15) with students & employees to build empathy and uncover deep-seated frustrations.
-  **Primary (Quantitative):** Google Form survey (n=250) to quantify pain points, travel habits, and validate the qualitative findings.

Secondary Research: Market Gaps

Public App-Based

Problem: No fixed parking. Fleets are chaotic, block pathways, and are unreliable. Business model is misaligned with campus needs (e.g., leaving campus).

Private Shuttles

Problem: Inflexible. They run on fixed routes and schedules, failing to provide the on-demand, point-to-point transport that users actually need.

Primary Research: Qualitative Insights



"I have 10 minutes to get from my lab to my lecture. The shuttle takes 15. Walking takes 20. I have to run, and I still show up late and sweaty."

— 3rd Year Engineering Student (Persona: "The Time-Pressed")

Primary Research: Current User Journey Map



Awareness

Class ends in 10 mins.
Next class is 1.5km away.

Anxiety, Stress



Consideration

Check shuttle app. Next
one is in 12 mins.

"Shuttle is useless"



Action

Starts speed-walking or
jogging across campus.

Physical strain



Arrival

Arrives at class 8 minutes
late, out of breath.

Embarrassment



Post-Action

Dreads the next cross-
campus class change.

Low satisfaction

Problem Synthesis & ROI



The Cost of Inefficiency

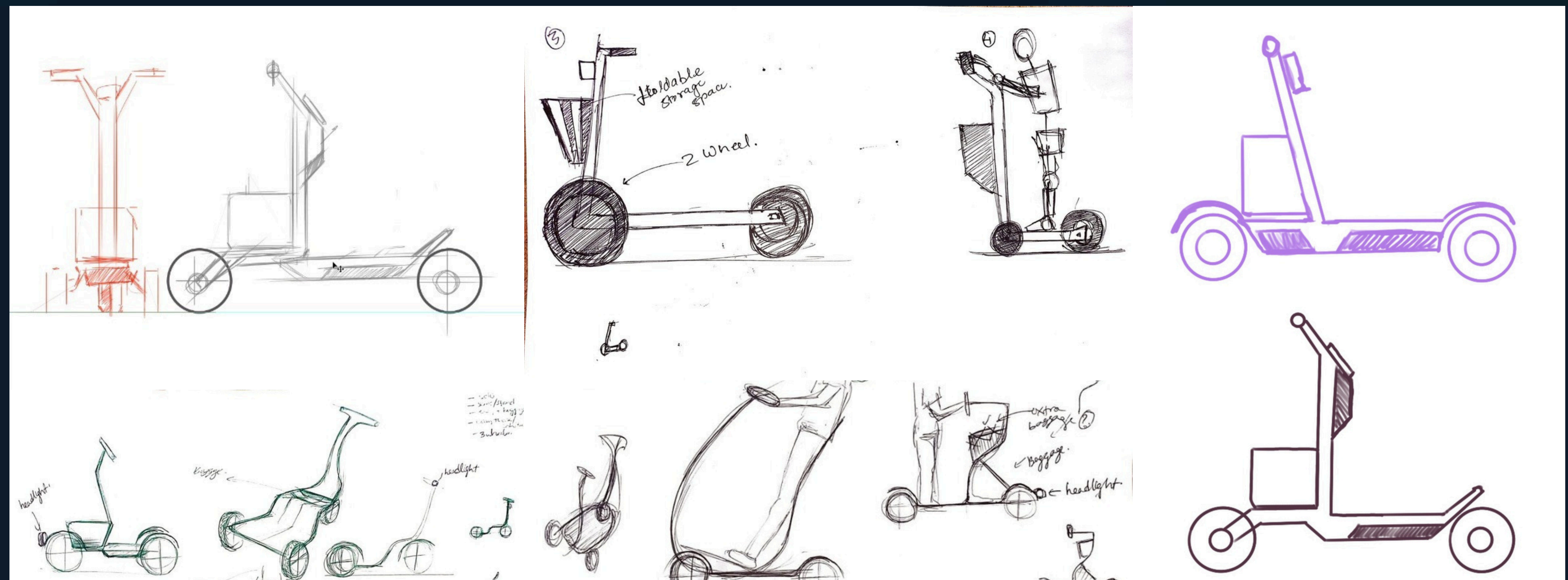
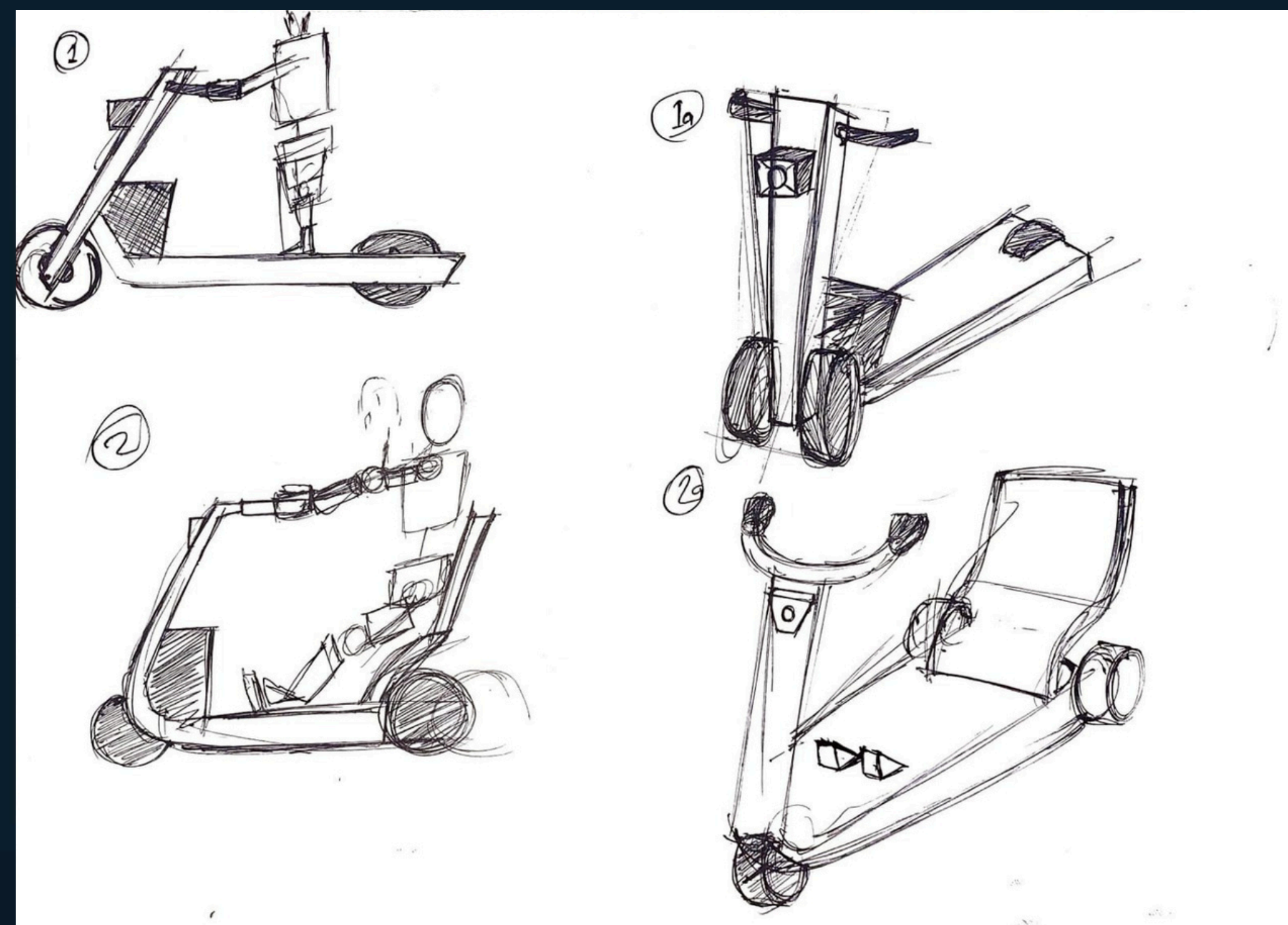
The problem is not just "inconvenience," it's a quantifiable waste of institutional resources.

- ✓ **Time Loss:**
 $(15 \text{ min walk}) \times (2.5 \text{ trips/day}) \times (20,000 \text{ students}) =$
12,500 hours
of wasted time per day.
- ✓ **Asset Underutilization:**
Students and faculty avoid useful buildings (libraries, labs) that are "too far away."
- ✓ **The Core Insight:**
A 5-minute ride replacing a 20-minute walk creates
15 minutes
of productive time, per user, per trip.

2. Develop

From research insights to a tangible, validated solution.

Concept Generation & Iteration



Concept Rejection Rationale

Criteria	Concept A (Pod)	Concept B (Trike)	Concept C (Reverse Triek)
Footprint (Width)	1.5m (Too wide)	1.1m (Too wide)	0.2 (Fits paths)
Pathway Compliance	Fails sidewalk & accessibility path standards.	Fails 0.9m minimum path standard.	Passes all path standards.
Cost (Est. BOM)	\$10,000+	\$3,000+	~\$1,200
Decision	Rejected	Rejected	Selected

The Selected Solution: A Closed-Loop Ecosystem



The Smart Vehicle

A commercial-grade vehicle with onboard IoT and a swappable battery, built for 24/7 uptime.



The Charging Hub

A smart parking and charging station that acts as the 'home' for every vehicle, eliminating chaos.



The Geo-Lock

An intelligent software fence that controls speed, access, and parking for total safety.

System Architecture



How It All Connects

The system is a constant loop of data.

- ✓ **1. User (ID) :**
Requests unlock at a Hub.
- ✓ **2. Cloud Server:**
Authenticates user, sends command to Hub & Vehicle.
- ✓ **3. Hub & Vehicle (IoT):**
Hub unlocks. Vehicle's IoT module activates, tracking GPS & speed.
- ✓ **4. Geo-Lock:**
Vehicle self-enforces "Slow-Go" or "No-Go" zones in real-time.
- ✓ **5. Parking:**
User docks at a new Hub. Hub confirms lock, ends ride, and reports status to Cloud.

Component 2: The Charging & Parking Hub

The End of Chaos

The hub is the anchor of the system. It's the only place a ride can be ended, ensuring vehicles are always charged and organized.

- Mechanical locking system
- Contact-based auto-charging
- LED status indicators
- Modular (2, 4, or 6-dock config)



Final Design Renders



Vehicle docked in the modular hub.

Rechargeable Battery

Handlebar UI and scan point.